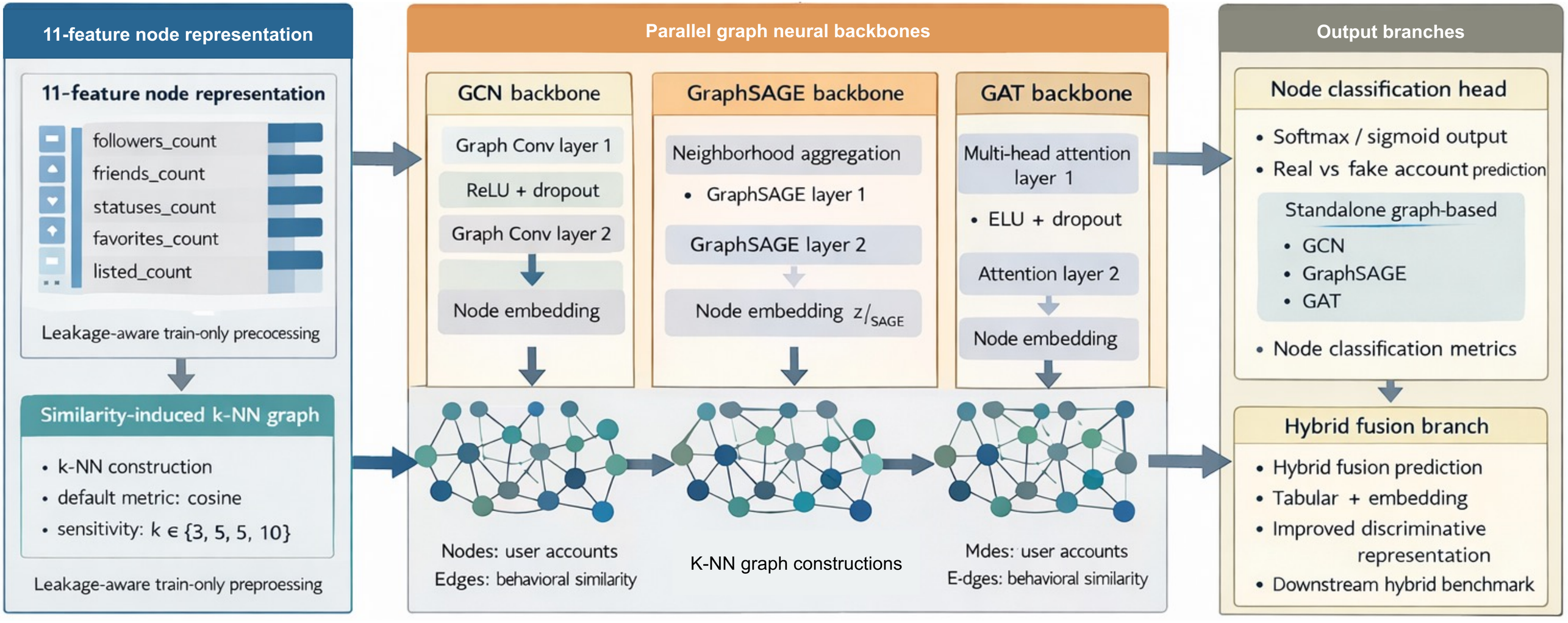
11-feature node representation
11-feature node representation
followers_count
friends_count
statuses_count
favorites_count
listed_count
Leakage-aware train-only precocessing
Similarity-induced k-NN graph
• k-NN construction
• default metric: cosine
• sensitivity: k ∈ {3, 5, 5, 10}
Leakage-aware train-only preproessing
Parallel graph neural backbones
GCN backbone
Graph Conv layer 1
ReLU + dropout
Graph Conv layer 2
Node embedding
GraphSAGE backbone
Neighborhood aggregation
• GraphSAGE layer 1
GraphSAGE layer 2
Node embedding z/SAGE
GAT backbone
Multi-head attention layer 1
• ELU + dropout
Attention layer 2
Node embedding
Nodes: user accounts
Edges: behavioral similarity
K-NN graph constructions
Mdes: user accounts
E-dges: behavioral similarity
Output branches
Node classification head
• Softmax / sigmoid output
• Real vs fake account prediction
Standalone graph-based
• GCN
• GraphSAGE
• GAT
• Node classification metrics
Hybrid fusion branch
• Hybrid fusion prediction
• Tabular + embedding
• Improved discriminative representation
• Downstream hybrid benchmark